Alan Zhou

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Education

Johns Hopkins University — Baltimore, MD
PhD in Cognitive Science (Computational Track) — *Aug. 2022 – July 2027*
Dissertation (expected July 2027): *An Information-Theoretic Account of Structural Disambiguation during Sentence Processing*
Advisor: John Hale

University of California, Berkeley — Berkeley, CA
B.A. in Computer Science and Cognitive Science — *Aug. 2017 – Dec. 2021*
Mentor: Gašper Beguš

Publications

** denotes equal contribution*

Peer-reviewed Journals and Conference Proceedings

- (2026) Alan Zhou*, Miloš Stanojević*, and John Hale. Syntactic Belief Update as the Driver of Garden Path Processing Difficulty. In *Proceedings of EMNLP 2026*. **PDF**
- (2026) Jane Li* and Alan Zhou*. (R)NNs too expressive? No relationship between neural architectures' theoretical expressivity and their empirical performance. In *Linguistics Vanguard*. **PDF**
- (2024) Alan Zhou and Colin Wilson. Modeling morphosyntactic agreement as neural search: a case study of Hindi-Urdu. In *Proceedings of the Society for Computation in Linguistics 2024*, pages 227–239. **PDF**
- (2023) Gašper Beguš*, Alan Zhou*, Peter Wu, and Gopala K Anumanchipalli. Articulation GAN: Unsupervised modeling of articulatory learning. In *ICASSP 2023 IEEE International Conference on Acoustics, Speech and Signal Processing*. **PDF** Video
- (2023) Gašper Beguš, Alan Zhou, and Christina Zhao. Encoding of speech in convolutional layers and the brain stem based on language experience. *Scientific Reports*. **PDF**
- (2022) Gašper Beguš and Alan Zhou. Interpreting intermediate convolutional layers of generative CNNs trained on waveforms. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30. **PDF**
- (2022) Gasper Begus and Alan Zhou. Modeling speech recognition and synthesis simultaneously: Encoding and decoding lexical and sublexical semantic information into speech with no direct access to speech data. In *Proc. Interspeech 2022*. **PDF**
- (2022) Gašper Beguš and Alan Zhou. Interpreting intermediate convolutional layers in unsupervised acoustic word classification. In *ICASSP 2022 IEEE International Conference on Acoustics, Speech and Signal Processing*. **PDF**

Presentations

- (2025) Jane Li and Alan Zhou. CNNs that robustly compute vowel harmony do not explicitly represent tiers. Poster at *Society for Computation in Linguistics*. **Abstract**
- (2025) Gašper Beguš, Thomas Lu, Alan Zhou, and Gopala Anumanchipalli. CiwaGAN: Informative imitation with articulatory learning. Poster at *Linguistic Society of America Annual Meeting*.
- (2025) Alan Zhou, Bruno Ferenc Segedin, Jingyi Chen, Sneha Ray Barman, and Gašper Beguš. Deep Language Learning: Modeling language from raw speech. Tutorial at *Linguistic Society of America Annual Meeting*. **Slides and Materials**

Awards and Honors

Dissertation Prize Fellowship Spring 2027
JHU Office of the Dean of Arts and Sciences $27,500 USD

Teaching

At Johns Hopkins

- **Foundations of Cognitive Science**, *Teaching Assistant* Fall 2025
- **Mathematical Foundations of Neural Network Theory**, *Teaching Assistant* Spring 2024
  - Led weekly recitation sessions offering help for homework and course content
  - Created and presented a set of three lectures introducing students to basic Python programming and constructing neural network models in PyTorch
  - Created a corresponding set of Python problem sets for each lecture.
- **Bayesian Inference**, *Teaching Assistant* Fall 2023, 2024
  - Presented a lecture on the Rational Speech Acts model of pragmatic inference (2024 offering)
- **Neuroscience: Cognitive**, *Teaching Assistant* Spring 2023

At UC Berkeley

- **Deep Learning and Phonology**, *Guest Lecturer* Fall 2021
  - Presented a guest lecture on high-performance computing to linguistics graduate students
- **Data Structures**, *Academic Intern* Spring 2018
  - Helped students in lab sections and office hours

Service

Department Lead, Early Career Colloquium Committee Fall 2024 to Present

- Led organization of a colloquium talk series meant to highlight the work of early postdocs and late PhD students across cognitive science
- Successfully secured funding from the Dean's office and coordinated visits

Member, General Colloquium Committee Fall 2025 to Present

- Helped coordinate student hosts for general colloquium visits

Skills

Programming Languages:	Python, Java, C, MATLAB, R, Lua, SQL
Tools/Technologies:	PyTorch, TensorFlow, Keras, Slurm, matplotlib Jupyter, Git, Gradle/Maven
Natural Languages:	English (fluent), Mandarin (conversational)